

Cryptography

MAT364 - Cryptography Course

Instructor: Adil Akhmetov

University: SDU

Semester: Fall 2025

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Course Description

The "**Cryptography for Programmers**" course is designed for students with computer science backgrounds and focuses on practical implementation of cryptographic algorithms in modern software systems.

Key Focus Areas

- **Practical Programming** of cryptographic algorithms
- **Real-world Applications** in software development
- **Minimal Mathematics** - more implementation details
- **Modern Tools** and libraries

Learning Outcomes

- Implement classical and modern ciphers
- Work with cryptographic libraries
- Create secure APIs and web applications
- Understand modern cryptographic protocols

Learning Objectives

Technical Skills

1. Practical Programming of Cryptographic Algorithms

- Implementation of classical and modern ciphers
- Working with cryptographic libraries
- Creating secure APIs and web applications

2. Understanding Modern Cryptographic Protocols

- TLS/SSL, HTTPS, SSH
- Authentication and authorization
- Digital signatures and certificates

Security Analysis

3. Security Analysis and Testing

- Penetration testing of cryptographic implementations
- Vulnerability analysis
- Secure programming practices

Course Structure

Weekly Course Plan

Weeks 1-4: Foundations

- **Week 1:** Introduction to Cryptography for Programmers
- **Week 2:** Classical Ciphers and Implementation
- **Week 3:** Cryptanalysis and Attacks
- **Week 4:** Stream Ciphers and One-Time Pads

Weeks 9-12: Applications

- **Week 9:** Digital Signatures
- **Week 10:** Cryptographic Protocols in Web Development
- **Week 11:** Cryptography in Mobile Applications
- **Week 12:** Cryptography in Blockchain

Weeks 5-8: Core Algorithms

- **Week 5:** Block Ciphers and Applications
- **Week 6:** Hash Functions and Data Integrity
- **Week 7:** Asymmetric Cryptography - RSA
- **Week 8:** Key Exchange and Protocols

Weeks 13-15: Advanced Topics

- **Week 13:** Quantum Cryptography and Post-Quantum
- **Week 14:** Practical Projects
- **Week 15:** Final Testing and Project Defense

Week 1: Introduction to Cryptography

Topics

- Cryptography fundamentals: terminology and concepts
- Types of attacks and security models
- Cryptographic primitives in programming

Practical Assignments

- Create a simple cipher in Python/JavaScript
- Analyze vulnerabilities in existing code
- Set up development environment for cryptography

Code Example

```
# Simple XOR cipher
def xor_cipher(text, key):
    return ''.join(chr(ord(c) ^ key) for c in text)

# Usage
message = "Hello World"
encrypted = xor_cipher(message, 5)
print(f"Encrypted: {encrypted}")
```

Week 2: Classical Ciphers Code Example

Topics

- Caesar cipher and its variations
- Substitution ciphers
- Transposition ciphers

```
def caesar_cipher(text, shift):  
    result = ""  
    for char in text:  
        if char.isalpha():  
            ascii_offset = 65 if char.isupper() else 97  
            result += chr((ord(char) - ascii_offset + shift  
        else:  
            result += char  
    return result
```

Practical Assignments

- Implement Caesar cipher with different keys
- Create frequency analysis program
- Break simple ciphers using brute force

Assessment Methods

Quizzes (60%)

- **3 Quizzes** throughout the semester
- Cover practical programming and theory
- Hands-on coding challenges
- Real-world problem solving

Final Project (40%)

- **Comprehensive cryptographic project**
- Choose from suggested topics or propose your own
- Implement a complete cryptographic system
- Documentation and presentation required

Suggested Final Projects: - Secure messaging application - File encryption system - Blockchain implementation - Web authentication system

Technical Requirements

Software

- **Python 3.8+** with libraries:
 - `cryptography`
 - `pycryptodome`
 - `requests`
- **Node.js** for web development
- **Git** for version control
- **Docker** for containerization

Recommended IDEs

- **Visual Studio Code** with cryptography extensions
- **PyCharm Professional**
- **IntelliJ IDEA**

Getting Started

```
pip install cryptography pycryptodome requests
npm install -g @slidev/cli
git clone [course-repository]
```

Reading List

Primary Literature

1. **"Real-World Cryptography"** - David Wong (2021)
2. **"Cryptography Engineering"** - Niels Ferguson, Bruce Schneier, Tadayoshi Kohno (2010)
3. **"Serious Cryptography"** - Jean-Philippe Aumasson (2017)

Online Resources

- OWASP Cryptographic Storage Cheat Sheet
- Cryptopals Crypto Challenges
- NIST Cryptographic Standards
- CryptoHack

Practice Tools

Network Analysis

Wireshark - Network traffic analysis - Protocol inspection - Security monitoring

Web Testing

Burp Suite - Web application testing - Vulnerability scanning - Security assessment

Password Security

John the Ripper - Password cracking - Security testing - Vulnerability assessment

Contact Information

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Important: This syllabus is adapted for students with computer science backgrounds and focuses on practical application of cryptography in modern software development.

Questions?

Thank you for your attention! ❤️